Topic: Annuities & Perpetuities **Name:** Solns

ANNUITIES

The original meaning of an annuity is that it is a fixed sum of money that is paid to a person (annuitant) each year at a specified time, usually for the rest of their lives.
The meaning today has changed and it covers many different situations where a regular equal payment or repayment is made and it does not necessarily have to be made annually. Some everyday examples of annuities in use are Variable Income Annuities, Fixed Annuities, Life Annuities, mortgage repayments, Equity Indexed Annuities, life insurance premiums and hire purchase repayments.

Ex 1.

At retirement Susie invests the $750 000 she has saved in her superannuation account in an account giving interest of 6% per annum compounded annually from which she will withdraw $60 000 after one year, and after each year thereafter. How much will be left in the account immediately after the tenth withdrawal (nearest dollar)?
If she wants the account to last her at least 35 years, withdrawing a constant amount each year, what should this constant amount be (rounded down to a multiple of $100)?

- Multiplier $= 100\% + 6\% = 106\% = 1.06$, as decimal

$$\begin{cases} T_{n+1} = 1.06\,T_n - 60000 \\ T_0 = 750\,000 \end{cases}$$

$\Rightarrow T_{10} = 552288.08$

$\approx \$552,288$ (nearest dollar) in the account "immediately after the tenth withdrawal".

OR

N = 10
I% = 6
PV = −750000
PMT = 60000
FV = ??
P/Y = 1
C/Y = 1

- Financial program

N = 35
I% = 6
PV = −750000
PMT = ?? ⇒ 51730.39 ≈ $51700 (rounded down to a multiple of $100)
FV = 0
P/Y = 1
C/Y = 1

Hence, Susie can withdraw up to $51,700 each year.

Solution

Initial amount invested:	\$750 000
Account balance after 1 year:	\$750 000 × 1·06 – \$60 000
Account balance after 2 years:	(\$750 000 × 1·06 – \$60 000) × 1·06 – \$60 000

We have $T_{n+1} = T_n \times 1{\cdot}06 - \$60\,000$,

and $T_0 = \$750\,000$

Using the recursion capability of some calculators, a spreadsheet or the inbuilt financial programs of some calculators we find that:

Immediately after the tenth withdrawal there will be \$552 288 in the account, to the nearest dollar.

☑ $a_{n+1}=1.06 \cdot a_n - 60000$
$a_0=750000$
☐ $b_{n+1}=\square$
$b_0=0$
☐ $c_{n+1}=\square$
$c_0=0$

n	a_n
6	6.5E +5
7	6.2E +5
8	6.0E +5
9	5.8E +5
10	$\boxed{5.5E+5}$

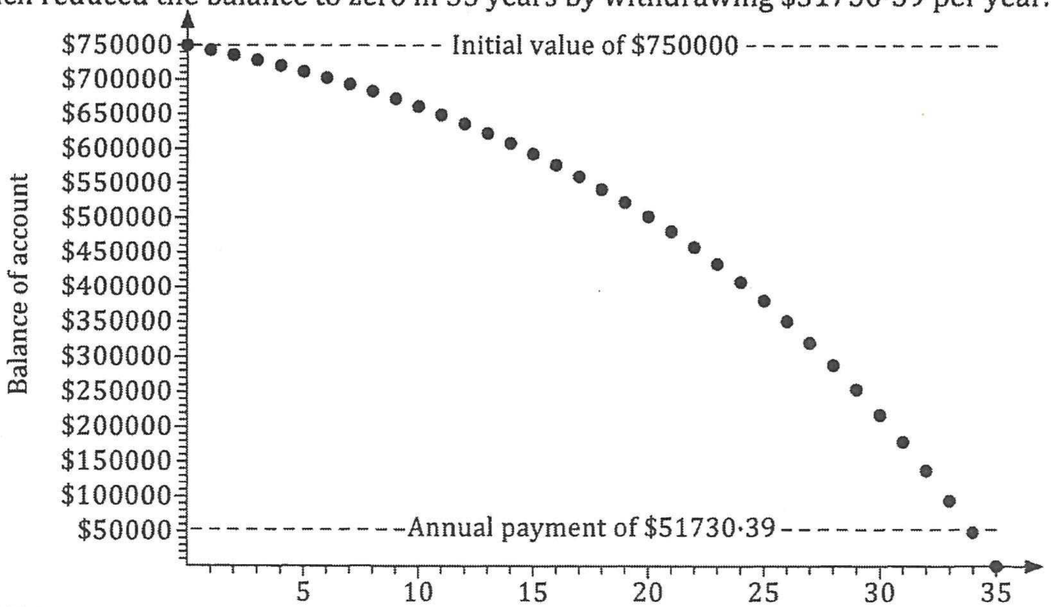
552288.075864291

Compound Interest

N	35
I(%)	6
PV	-750000
Pmt	51730.39423
FV	0
PpY	1
CpY	1

Varying the withdrawal amount to reduce the balance to zero in 35 years, or using a calculator with an inbuilt financial capability, we can determine that Susie can withdraw up to \$51 700 each year (rounded down to a multiple of \$100) for the account to last at least 35 years.

The graph below shows the declining balance in the account of the previous example which reduced the balance to zero in 35 years by withdrawing \$51730·39 per year.



Can your calculator display a graph of this type from the terms of a recursively defined sequence? Investigate.

Annuities.

In the previous example Susie managed her own *income stream* by making a deposit into an account, withdrawing a fixed amount each year to live on and allowing the balance of the account to earn interest. A compound interest investment from which regular payments are made for a fixed period of time is called an **annuity**.

Making regular payments to pay off a loan, as encountered in the previous chapter, is a bit like an annuity. The finance company invests money with us in the form of a loan and we make regular payments back. However the word annuity is usually used in the context of using a sum of money to set up a regular income.

By making a one off payment a person can enter into an annuity arrangement whereby they will receive a guaranteed fixed regular amount for an agreed period of time. Such arrangements can be very useful for generating a steady income in retirement

Fixed term annuities make regular payments for a fixed period of time. If the person who purchased the annuity dies before the end of the fixed period of time the remaining payments, or the remaining balance, would pass to a beneficiary.

Ex 2.

Richard invests $450 000 in an annuity at a fixed rate of 8% per annum compounded annually. He receives a payment of $50 000 at the end of each year after interest for the year has been added to his account.
(a) What will be the balance of Richard's account after 10 years?
(b) How long will the annuity last?
(c) What amount could be paid at the end of each year so that the value of the investment remains at $450 000?

a) multiplier = 100% + 8% = 108% = 1.08 (decimal)

"Financial"
N = 10
I% = 8
PV = -450000
PMT = 50000
FV = ??
P/Y = 1
C/Y = 1

$$\begin{cases} T_{n+1} = 1.08 T_n - 50000 \\ T_0 = 450000 \end{cases}$$

$\Rightarrow T_{10} = \$247188.13$

Hence, the balance after 10 years is $247,188.13

b) $T_{16} = 25460.04$
& $T_{17} = -22503.16$

$\Rightarrow$ The annuity will last for 17 years.

& last payment = 50000 - 22503.16 = $27,496.84

c) For the investment to remain at $450,000, we withdrawn the interest only at the end of each year.

Interest $= \frac{8}{100} \times 450000$

$= \$36,000$

Ex 3. [Ex4A, Q6] < "Financial" "Sequence"

An insurance company offers an annuity that pays 0.5% per month with a monthly withdrawal facility at the end of every month on deposit of invested amount. . Eric invests $500 000 with this company and decides to withdraw $5000 at the end of each month.

(a) How many withdrawals of $5000 will Eric make before his account balance falls below $5000?

(b) What will be the amount of Eric's final withdrawal?

(c) How long will Eric's investment last if he made withdrawals of $4000 each month?

(d) What will be the amount of Eric's final withdrawal?

(e) For how long would Eric be able to withdraw $5000 per month if the interest rate was 7% per annum?

(f) For how long would Eric be able to withdraw $4000 per month at a rate of 7.5% p.a.?

a) method 1

N = ?? ⇒ 137.97

I% = 6%

PV = -500 000

PMT = 5000

FV = 5000

P/Y = 12

C/Y = 12

Hence, 138 withdrawals.

method 2: sequence

$T_{n+1} = 1.005 - 5000$

$T_0 = 500\,000$

$T_{137} = 9805.60$

$T_{138} = 4854.63$ *

Hence, 138 withdrawals

b) $T_{139} = -121.10$ | 4878.903194 ≈ $4878.90

Hence, last withdrawal

= 5000 − 121.10 = $4878.90

c) $\begin{cases} T_{n+1} = 1.005 - 4000 \\ T_0 = 500\,000 \end{cases}$

$T_{196} = 2612.62$

$T_{197} = -1374.32$

Hence, 197 months

d) Final withdrawal = 4000 − 1374.32

= $2625.68

e) 7% pa = $\frac{7}{12}\% = 0.58\dot{3}\% = 0.0058\dot{3}$

Multiplier $1 + 0.0058\dot{3} = 1.0058\dot{3}$

$\begin{cases} T_{n+1} = 1.0058\dot{3}\, T_n - 5000 \\ T_0 = 500\,000 \end{cases}$

$T_{150} = 2576.92$

$T_{151} = -2408.04$

Hence, 150 withdrawals of $5000/month

f) $\frac{7.5\%}{12} = 0.625\% = 0.00625$

Multiplier $1 + 0.00625 = 1.00625$

$\begin{cases} T_{n+1} = 1.00625\, T_n - 4000 \\ T_0 = 500\,000 \end{cases}$

$T_{243} = 3702.63$

$T_{244} = -274.23$

Hence, 243 withdrawals of $4000/month.

Complete Ex 4A

Using the Financial Application of a graphic calculator

Ex 4.

Renata invests $60 000 in an annuity at a rate of 7% per annum compounded annually. She receives a payment of $10 000 every year after interest for the year has been added to her account. How long, to the nearest year will the annuity last?

"Financial"

N = ?? ⇒ 8.051109322

I% = 7

PV = -60000

PMT = 10000

FV = 0

P/Y = 1

C/Y = 1

Hence, the annuity lasts 8 years (to the nearest year)

Ex 5.

To ensure a steady income during his retirement Kevin invests the $480 000 he has in his superannuation account in an annuity at a rate of 6.7% per annum compounded monthly. What amount should Kevin withdraw from his annuity investment each month if he wishes to receive payments for 15 years?

"Financial"

N = 180 ← 15×12

I% = 6.7

PV = -480000

PMT = ?? ⇒ 4234.270451

FV = 0

P/Y = 12

C/Y = 12

Hence, Kevin should withdraw $4234.27 per month.

Complete Ex 4B

Indexing

In some cases the regular amount paid out from an annuity might increase over time so that the amount received keeps up with inflation and maintains its "buying power". We say that the regular payments are *indexed*, i.e. the payments change in line with the movement of some price index, eg the consumer price index, a measure of the increase in the price of a typical household selection of goods.

Ex 6.

A person wishes to deposit \$200000 into an account that will earn interest of 8% per annum, compounded annually, withdrawing \$25000 at the end of the first year, \$25500 at the end of the second year, \$26010 at the end of the third year, and so on with each annual withdrawal being a 2% increase on the withdrawal of the previous year.

(a) What will be the balance of the account immediately after the fifth withdrawal?

(b) How long will it take for the balance in the account to reduce to zero?

Initial \$200,000

After 1 year: $200,000 \times 1.08 - 25,000 = \$191,000$

2 years $191000 \times 1.08 - 25000 \times 1.02 = \$180780$

⋮

$$\begin{cases} T_{n+1} = T_n \times 1.08 - 25000 \times 1.02^n \\ T_0 = 200,000 \end{cases}$$

$\Rightarrow T_5$, the account balance after 5 years (after the fifth withdrawal)

$= \$141,679.25$

b)

$T_{11} = 12884.18$

$T_{12} = -17169.44$

Hence, the balance will reduce to zero at the end of the 12th year.

[The final withdrawal $= 12884.18 \times 1.08 \simeq \$13914.91$]

PERPETUITIES

A **perpetuity** is an annuity in which regular payments are paid from the investment that continues forever. For regular payments to be paid forever from an investment implies that the value of the investment remains constant. For this to be so the value of the regular payment must be equal to the interest earned on the investment.

Perpetuities are sometimes referred to as **perpetual annuities**.

Example 1:

Suppose we invest \$5000 in an account paying 8% per annum compounded annually, and we withdraw \$400 at the end of each year.

Initial amount invested:	\$5000		
Account balance after 1 year:	$\$5000 \times 1{\cdot}08 - \$400$	I.e. \$5000.	
Account balance after 2 years:	$\$5000 \times 1{\cdot}08 - \$400$	I.e. \$5000.	
Account balance after 3 years:	$\$5000 \times 1{\cdot}08 - \$400$	I.e. \$5000.	Etc.

In this case the annual withdrawal of \$400 could continue forever because each year the interest earned exactly equals the amount paid out. This sort of annuity is called a **perpetuity**, the word perpetuity meaning *lasting forever*.

Example 2:

Consider an investment of \$2000 invested at 5% per annum compounded annually with annual withdrawals of \$100. To find value of this annuity some years into the future we can write a recurrence relation.

That is: $T_{n+1} = 1.05T_n - 100, \ T_0 = 2000$

Now the present value is:	2000
Value after 1 year:	1.05 x 2000 – 100 = 2100 – 100 = 2000
Value after 2 years:	1.05 x 2000 – 100 = 2100 – 100 = 2000
Value after 3 years:	1.05 x 2000 – 100 = 2100 – 100 = 2000

The annual withdrawal of \$100 could continue indefinitely, this is because the interest earned each year (0.05 x 2000 = 100) is exactly the same amount as the withdrawal.

This annuity is a perpetual annuity or a perpetuity.

General Case:

Now consider the general case where: P is the present value or the amount invested
r is the percentage interest rate per period written as a decimal
Q is the value of the regular payment per period

Then:

$(1 + r) \times P - Q = P$	
$P + Pr - Q = P$	Expanding the parentheses
$Pr - Q = 0$	Subtracting P from both sides
$Pr = Q$	Adding Q to both sides

Hence $Q = Pr$

$$Q = P \cdot r$$

Regular payment

Present Value

Interest rate (decimal

Ex 7.

The director of music at a particular college wishes to set up a scholarship where every year a worthy student receives $2000 to assist with music tuition fees. How much needs to be invested at 8% per annum compounded annually to set up this perpetuity?

Method 1

$$(1+0.08) \times P - 2000 = P$$

Solver:

$$P = 25000$$

Hence, $25,000 needs to be invested.

Method 2

$$Q = Pr$$

$$2000 = P \times 0.08$$

$$P = \frac{2000}{0.8} = 25000$$

$\therefore$ $25,000 needs to be invested.

Ex 8.

Helen invests \$400 000 in a perpetuity which will provide a monthly income without using any of the invested amount of \$400 000. If the interest rate on the perpetuity is 8.4% per annum compounded monthly, what monthly payment will Helen receive?

<u>Method 1</u>

$$\left(1+\frac{0.084}{12}\right)\times 400000 - Q = 400000$$

Solver: $Q = 2800$

$\therefore$ Helen will receive a monthly payment of \$2,800

<u>Method 2</u>

$$Q = P.r$$

$$Q = 400000 \times \frac{0.084}{12}$$

$$= 2800$$

$\therefore$ Helen will receive a monthly payment of \$2800.

Complete Ex 4C

INCREASING ANNUITY INVESTMENTS

An increasing annuity investment is an investment that involves making an initial deposit followed by additional regular, fixed deposits (or payments) into an interest earning account earning a fixed rate of compound interest.

Ex 9.

On the birth of their grandson his grandparents invest \$50 in a term deposit account. Every year on his birthday, they add \$50 to the initial investment and renew the account. The investment earns interest at the rate of 7% compound interest adjusted annually.

(a) How much will the grandson have in this account on his 17th birthday, if \$50 is added to the investment on that day?

(b) How much interest was earned by this investment?

a) $\begin{cases} T_{n+1} = 1.07\,T_n + 50 \\ T_0 = 50 \end{cases}$

T_{17} = (his 17th birthday)

$T_{17} = \$1699.95$

or "Financial"

N = 17

I% = 7

PV = -50

PMT = -50

FV = ?? ⇒ 1699.951626

b) Interest = 1699.95 − 18 × 50

= \$799.95

SOLUTION

(a) The spreadsheet below shows the progress of the investment from the initial deposit of $50 at birth of the grandson to the balance on his 17th birthday after the $50 has been added on that day.

"too long" ⇒ time consuming

	A	B	C	D	E	F
1	Birthday	Balance at start	Interest	Deposit	Balance at end	
2	0	0.00	0.00	50.00	50.00	The formulae used
3	1	50.00	3.50	50.00	103.50	to generate this
4	2	103.50	7.25	50.00	160.75	spreadsheet are
5	3	160.75	11.25	50.00	222.00	listed below:
6	4	222.00	15.54	50.00	287.54	
7	5	287.54	20.13	50.00	357.66	Cell E2 =B2+C2+D2
8	6	357.66	25.04	50.00	432.70	
9	7	432.70	30.29	50.00	512.99	Cell B3 =E2
10	8	512.99	35.91	50.00	598.90	
11	9	598.90	41.92	50.00	690.82	Cell C3 =0.07*B3
12	10	690.82	48.36	50.00	789.18	
13	11	789.18	55.24	50.00	894.42	Cell D3 =D$2
14	12	894.42	62.61	50.00	1007.03	
15	13	1007.03	70.49	50.00	1127.52	
16	14	1127.52	78.93	50.00	1256.45	
17	15	1256.45	87.95	50.00	1394.40	
18	16	1394.40	97.61	50.00	1542.01	
19	17	1542.01	107.94	50.00	1699.95	

Cell E19 gives the amount in the account on the boys 17th birthday.
Therefore we can see that there will be $1699.95 in the account on his 17th birthday.
(b) Interest earned may be determine by finding the sum of the entries in column C using the command "=sum(C2:C19)" which will give the amount $799.95

Alternatively, the problem may be solved using a recurrence relation by finding a sequence which identifies the amount in the investment account on the grandson's birthday.

(a)
Amount at birth:	**50**
Amount of 1st birthday:	**50**(1.07) + 50
Amount of 2nd birthday:	**(50(1.07) + 50)**(1.07) + 50
Amount of 3rd birthday:	**[(50(1.07) + 50)(1.07) + 50]**(1.07)+ 50

With $T_1 = \mathbf{50}$, $T_2 = \mathbf{50}(1.07) + 50$, $T_3 = \mathbf{(50(1.07) + 50)}(1.07) + 50$, $T_4 = \mathbf{[(50(1.07) + 50)(1.07) + 50]}(1.07) + 50$ etc, we have established a pattern or sequence that can be defined using a recurrence relation.

That is: $T_{n+1} = 1.07T_n + 50$, $T_1 = 50$ where T_1 is the amount at birth and hence T_2 is the amount in the account on the first birthday, T_3 is the amount in the account on the second birthday, etc.

It makes sense to redefine the recurrence relation so that the amount on a particular birthday is the same as the term number, hence
$T_{n+1} = 1.07T_n + 50$, $T_0 = 50$ where T_0 is the amount at birth and T_1 is the amount in the account on the first birthday, T_2 is the amount in the account on the second birthday, and so on.
On entering the recurrence relation in the sequence application of a calculator, the progress of this investment can be viewed and analysed.
Hence the amount in this account on his 17th birthday will be given by $T_{17} = 1699.9516\ldots$, that is $1699.95.

(b) The interest earned by the investment can be easily found by finding the difference between the final value of the investment on the grandson's 17th birthday and the total invested by the grand parents

That is: Interest earned = Final value of the investment – Total invested
= $1699.95 – 18 x $50
= $799.95

Notes:

When solving increasing annuity investment problems it is very important that each step be worked from first principles so that a pattern can be established by considering each of the following:

(i) what interest rate must be applied?
(ii) when is the interest paid?
(iii) when are payments made?
(iv) what is the first term of the sequence?
(v) when does the annuity mature?

Once a pattern has been established then a spreadsheet or recurrence relation may be applied to solve the problem.

* **Ex 10.** (P/Y and C/Y are different!)

Suppose that $250 000 is invested into an account paying 8% interest, compounded quarterly, with $40 000 withdrawn from the account at the end of the first year and at the end of every year thereafter.
Determine the balance of the account immediately after the sixth withdrawal.

Initial, $\$250,000$

After 1 year: $250000 \times 1.02^4 - 40000 = 230608.04$

" 2 years: $230608.04 \times 1.02^4 - 40000 = 209617.56$

⋮

$T_{n+1} = T_n \times 1.02^4 - 40000$

$T_0 = 250000$

Balance after the 6th withdrawal:

$T_6 = \$106866.65$

OR "Financial"

$N = 6$
$I\% = 8$
$PV = -250000$
$PMT = 40000$
$FV = ?? \Rightarrow 106866.6549$
$P/Y = 1$
** $C/Y = 4$

Ex 11.

Suppose that \$250 000 is invested into an account paying 8% interest, compounded annually, with \$10 000 withdrawn from the account three months later and a further \$10 000 each quarter thereafter.
Determine the balance of the account after three years.

"Financial"

$N = 12$

$I\% = 8$

$PV = -250000$

$PMT = 10000$

$FV = ?? \Rightarrow 181238.775$

** $P/Y = 4$

$C/Y = 1$

The balance after 3 years is \$181,238.78

$x^4 = 1.08 \Rightarrow x = 1.019426547$

OR

$T_{n+1} = 1.019426547\, T_n - 10000$

$T_0 = 250,000$

$\Rightarrow T_{12} = 181238.78$

Complete Ex 4D

Using spreadsheets

Ex 12.

David decides to save a fixed sum of money every year as an emergency fund for any unforeseen expenses. He decides to open an investment account with a deposit of $500 on the 1st January 2015 and then deposit the same sum on the 1st January each year. To keep track of his investment David set up a apreadsheet and the portion below shows the progress of his account for the first three years. Entries have been rounded to two decimal places.

	A	B	C	D	E	E
1	Date	Deposit	Balance at start	Interest	Balance at end	
2	1st Jan 2015	500	500.00	40.00	540.00	
3	1st Jan 2016	500	1040.00	83.20	1123.20	
4	1st Jan 2017	500	1623.20	129.86	1753.06	

(a) Calculate the annual interest rate on David's investment.

(b) Write a recurrence relation that gives the balance B_n of David's account at the end of the n^{th} year.

(c) Find the amount of his investment at the end of 2025.

(d) Find the interest earned by the end of 2025.

a) Interest $= \frac{40}{500} \times 100\% = 8\%$ p.a.

b) Initial : \$500

After 1 year : $500 \times 1.08 = 540$

2 years $(540 + 500) \times 1.08 = 1123.20$

⋮

B_n is the balance at the end of the nth year:

$$\begin{cases} B_{n+1} = 1.08(B_n + 500) \\ B_1 = 540 \end{cases}$$

or

$$\begin{cases} B_n = 1.08(B_{n-1} + 500) \\ B_1 = 540 \end{cases}$$

c) 2025 ⇒ 11th term ⟶ $B_{11} = \$8988.56$

Amount of the investment at the end of 2025 is \$8988.56

d) Interest $= 8988.56 - 500 \times 11$

$= \$3488.56$

$\frac{1.5}{200} = 0.0075$

Ex 13. [Ex4E, Q5]

A self-employed tradesman invests $200 on the first day of each month into a fund as a nest egg for his retirement. The spread sheet below shows part of this account.

	A	B	C	D	E	F
1	Month	Deposit	Balance at start of the month	Interest	Balance at end of the month	
2	1	200	200.00	1.50	201.50	
3	2	200	401.50	3.01	404.51	
4	3	200	604.51	4.53	609.05	
5	4	200	809.05	6.07	815.12	

$\times 0.0075$ $C+D$

(a) Calculate the annual interest rate for this investment account.

$\frac{1.50}{200} \times 100 = 0.75\%/\text{month}$ (0.0075 in decimal)

$\Rightarrow 0.75\% \times 12 = 9\%$ per year

(b) Complete row 5 of the spreadsheet.

(c) What is the value of this investment after 10 years?

$\begin{cases} B_{n+1} = 1.0075(B_n + 200) \\ B_1 = 201.5 \end{cases}$

After 10 years $\Rightarrow T_{120} = \$38{,}993.13$

(d) How much interest has this investment accumulated over the 10 years

Interest $= 38993.13 - 200 \times 120$

$= \$14{,}993.13$

(e) If the tradesman decided to double his deposits, would this action double the amount in the account at any given time? Discuss, support your answer mathematically.

$B_{n+1} = 1.0075(B_n + 400)$

$B_1 = 403$

400×1.0075

After 10 years $\Rightarrow T_{120} = 77986.25$

comparing $38993.13 \times 2 = 77986.26$

$\Rightarrow$ only 1 cent difference due to rounding.

$\therefore$ Yes, doubling the deposit will double the value of the investment.

(f) If the interest rate was doubled, would this action double the amount of interest earned? Discuss, support your answer mathematically.

$\frac{18\%}{12} = 0.015$ (decimal)

month 1: $200 \times 0.015 = 3$

$403 \times 0.015 = 6.045$ } more than double

$\therefore$ No, if the interest rate was doubled, the amount of interest is doubled for the 1st month only, but for subsequent months the interest is more than double.

(g) If the tradesman had initially negotiated a interest rate of 12.6% per annum how much extra would his investment be worth after 10 years?

$\frac{12.6\%}{100} = 0.126$; $\frac{0.126}{12} = 0.0105$

$\begin{cases} B_{n+1} = 1.0105(B_n + 200) \\ B_1 = 202.1 \end{cases}$

After 10 years: $T_{120} = 48164.06$

Extra $= 48164.06 - 38993.13$

$= \$9{,}170.93$

Complete Ex 4E

NUMBER OF PAYMENTS PER YEAR ≠ NUMBER OF TIMES INTEREST IS COMPOUNDED PER YEAR

Our discussion of annuities to date has involved situations where the number of payments per year corresponded to the number of times interest was compounded per year. In real life situations the frequency of payments per year may not be the same as the frequency of compounding per year.
In this section of this chapter we shall consider situations where these frequencies are different.

Frequency of compounding greater than the number of payments

Ex 14.

C/Y = 12

Alba invests $360 000 in an annuity at a rate of 9% per annum compounded monthly. She receives a regular payment of $45 000 at the end of every year. → P/Y = 1

(a) What is the balance of the annuity after 5 years?
(b) How long, to the nearest year will the annuity last?
(c) What is amount of the last payment?

a) "Financial" program

N = 5
I% = 9
PV = −360 000
PMT = 45000
FV = ?? ⇒ 292282.9688
P/Y = 1
C/Y = 12

Hence, the balance after 5 years is $292,282.97

OR

$\frac{0.09}{12} = 0.0075$

⇒ multiplier = 1.0075

Initial 360 000

After 1 ~~month~~: 360000 (1.0075)

After 2 months: $360000 (1.0075)^2$

⋮

⇒ After 12 months = $360000 \times 1.0075^{12}$

Hence, Amount invested = 360000

After 1 year: $360000 \times 1.0075^{12} - 45000$

⋮

$$\begin{cases} T_{n+1} = 1.0075^{12} T_n - 45000 \\ T_0 = 360\,000 \end{cases}$$

b) ⇒ N = 15.48128557

Hence, the annuity lasts 16 years

c) last payment = 45000 − 22838.30 ← use N = 16 ⇒ FV = 22838.30

= $22,161.70

Number of payments per year greater than the frequency of compounding

Ex 15. P/Y = 4 C/Y = 1

Donald invests $600 000 in an account which pays 8.7% per annum compounding annually. He withdraws $20 000 each quarter for living expenses.

(a) What is the balance of the account after 5 years?

(b) How long does it take to reduce the balance of the account to $0?

a) "Financial" program

N = 20

I% = 8.7

PV = -600000

PMT = 20000

FV = ?? ⇒ 419359.5177

P/Y = 4

C/Y = 1

Hence, the balance of account after 5 years is $419,359.52

b)

FV = 0

⇒ N = 47.9637 1801

⇒ 48 withdrawals

It takes 12 years ($\frac{48}{4} = 12$) to reduce the balance of the account to $0.

Ex 16. [Ex 4F, Q18]

Gerry invests \$450 000 in an annuity at a fixed rate of 6.6% per annum compounded monthly. He receives a payment of \$34 000 at the end of each half year.

(a) How many payments of \$34 000 will Gerry receive?

(b) What will be the final payment he receives that closes the account?

a) "Financial"

$N = ?? \Rightarrow 17.77163165$

$I\% = 6.6$

$PV = -450000$

$PMT = 34000$

$FV = 0$

$P/Y = 2$

$C/Y = 12$

Hence, 17 payments of \$34,000.

b) $\therefore N = 18$

$\Rightarrow FV = -7666.23$

Hence, the final payment $= 34000 - 7666.23$

$= \$26{,}333.77$

Complete Ex 4F